

The Absorption Coefficient of an Unknown Material

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Introduction and Aim

A source of (^{60}Co) emits gamma rays and some of these gamma rays can be shielded away from an observation point by an “absorber” with a specific linear absorption coefficient (a measure of how well the absorber absorbs a beam of radiation per unit thickness of absorber (Medical Dictionary, n.d.)). The linear absorption coefficient of a material is dependent on the density of the material of the absorber (with a greater density allowing for more absorption) (Wasan, Jassim, & Ammar, 2019). The linear absorption coefficient, α , satisfies $N(x) = N_0 e^{-\alpha x}$ where N_0 is the radiation measured from the source with no absorber present and $N(x)$ is the observed radiation for thickness x . Thus, the aim of this is to use this formula to calculate the linear absorption coefficient of an unknown material which has the same density as lead, but is different to lead, and show this coefficient is the same as the coefficient of lead.

Method and Apparatus

Apparatus

- A Geiger Machine, used to measure the counts (a measure of radiation) emitted through the absorber over a period of time
- 4 Slabs of the same unknown material (thickness does not matter). This is to get a range of values to plot to work out the linear absorption coefficient
- A slab of lead
- A ruler, to measure the thickness of the slabs and slab of lead
- A radioactive source of (^{60}Co)

A diagram of the apparatus used

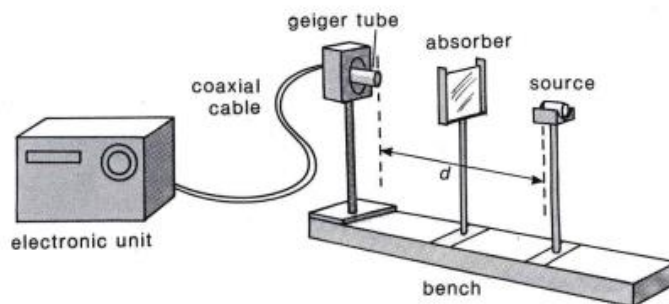


Diagram 1

Method

First, the Geiger Machine was configured by switching it on, waiting until it was set up, then pressing SELECT and changing the “Adjust Gate Time” setting to 60 seconds, then pressing SELECT to confirm this. Note if the voltage is not 550V, it must be configured to this first before pressing SELECT to change the time. Next, the natural radiation was observed by pressing the START-STOP dial. After 60 seconds, the displayed number of counts was recorded. This was repeated four times to get 4 readings. Next, the source was set at some distance from the Geiger tube (this is not important, must just remain fixed). The START-STOP dial was pressed to get a reading with no absorber. This was done 4 times and each recording was recorded. Next, one slab was taken and the length of this was measured using a ruler. The slab of the absorber was placed between the source and Geiger tube (as displayed above) and 4 readings with that slab were recorded. Another slabs thickness was measured and then placed next to the first absorber and 4 readings were taken again. This was done for all 4 slabs. After this, the slabs were removed, and the slab of lead was placed. The lead slab’s thickness was measured and the counts with this slab as an absorber were measured to get 4 readings.

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Data and Results

Tables and Graphs

Table 1: Showing the raw data for the method above

A table showing the 4 readings for the counts of each reading along with the thickness (mm) of the slabs used

Number of slabs	Thickness of the Slabs used (mm)	Count readings for that thickness
0	No slabs	193; 190; 176; 184
1	2	127; 129; 127; 137
2	2; 3	92; 119; 115; 117
3	2; 3; 2	109; 101; 99; 103
4	2; 3; 2; 3	92; 89; 91; 80

Now we need to consider the natural background. The counts for the natural radiation was measured at: 22; 22; 18; 13. Thus $\overline{N}_{nat} = \frac{22+22+18+13}{4} = 18.8$ counts per minute. So, by averaging the thickness and the counts and subtracting this natural background, we get:

Table 2: Showing the raw data from Table 1 after averaging and accounting for the natural background

A table showing the average number of counts of each reading along with the total thickness (mm) of the slabs used

Number of slabs	Thickness of the Slabs used (mm)	Count readings for that thickness (per minute)
0	0	167
1	2	111
2	5	92.0
3	7	84.2
4	10	69.2

Example Calculation with 0 slabs: $\overline{N}_0 = \frac{193+190+176+184}{4} = 185.75$, so $N_0 = 185.75 - 18.8 \approx 167$

For the lead block, the counts were measured as: 86; 67; 84; 75

Now the best approximation for lead is: $N_{lead} = \frac{86+67+84+75}{4} - 18.8 = 59.2$ counts.

The length of the lead block was measured to be 12mm

Note that $N_{lead} = N(12)$ for the lead block

Our initial equation was: $N(x) = N_0 e^{-\alpha x} \Leftrightarrow e^{-\alpha x} = \frac{N(x)}{N_0} \Leftrightarrow -\alpha x = \ln\left(\frac{N(x)}{N_0}\right)$. Now we calculate $\ln\left(\frac{N(x)}{N_0}\right)$ for each x

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Table 3: The data from Table 2 used to get $\ln\left(\frac{N(x)}{N_0}\right)$ for each x

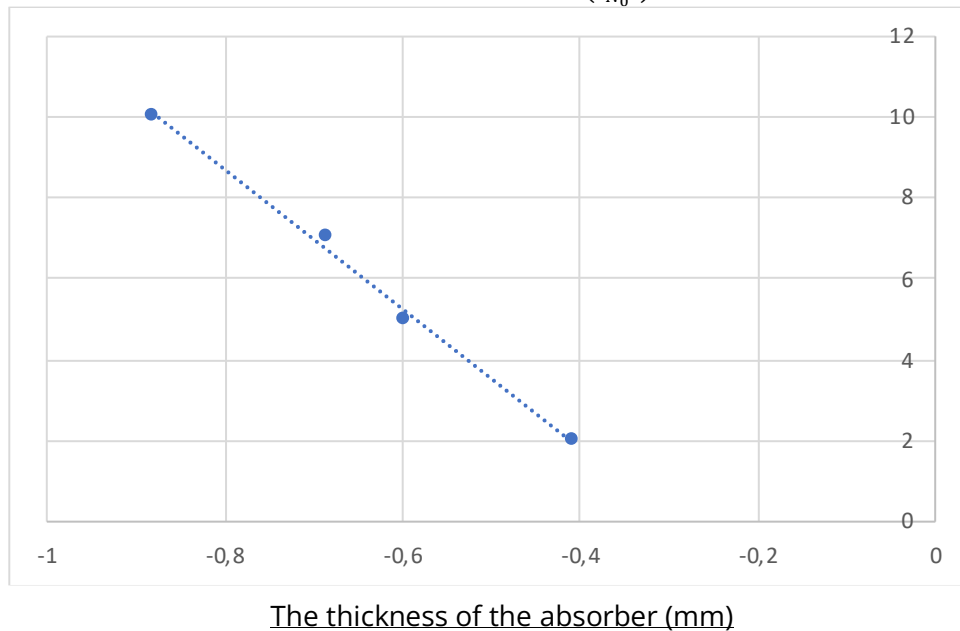
Table showing $\ln\left(\frac{N(x)}{N_0}\right)$ and x used to plot Graph 1

x (mm)	$\ln\left(\frac{N(x)}{N_0}\right)$
2	-0.408
5	-0.596
7	-0.685
10	-0.881

Graph 1: Plotting the data from Table 5 to get a line of best fit and calculate the linear absorption coefficient

Depicting the linear relationship between $\ln\left(\frac{N(x)}{N_0}\right)$ and the thickness (mm) of an absorber

$\ln\left(\frac{N(x)}{N_0}\right)$ for each an absorber with thickness x (mm)



Analysis and Discussion

Calculating with Graph 1

Our initial equation was: $N(x) = N_0 e^{-\alpha x} \Leftrightarrow e^{-\alpha x} = \frac{N(x)}{N_0} \Leftrightarrow -\alpha x = \ln\left(\frac{N(x)}{N_0}\right)$

This is in the form $Y=mX$ which is why we plot the data in Table 3 to get Graph 1

Using the least squares method on Graph 1, we get:

$$m = -0.0583 \pm 0.0025$$

$$c = -0.2929 \pm 0.0166$$

So $m = -\alpha \Rightarrow \alpha = -m = 0.0583 \text{ mm}^{-1}$ and $u(\alpha) = \alpha \sqrt{\left(-\frac{u(m)}{m}\right)^2} = 0.0583 \sqrt{\left(-\frac{0.0025}{0.0583}\right)^2} = 0.0025$

So $\alpha = (0.05830 \pm 0.00250) \text{ mm}^{-1}$

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Calculating with Lead Block

Rearranging the equation gives $\alpha = -\frac{\ln\left(\frac{N(x)}{N_0}\right)}{x}$

Firstly, there is an uncertainty in the length reading. Since this is a triangular PDF,

$$u(x_{reading}) = \frac{12.5-11.5}{2\sqrt{6}} = 0.204\text{mm}$$

Now on the lead block for $x = 12$, $N(x) = N_{lead} = 59.2$ and $N_0 = 167$ so

$$\alpha = -\frac{\ln\left(\frac{59.2}{167}\right)}{12} = 0.0864$$

To calculate the uncertainty, first set $A = \ln\left(\frac{N(x)}{N_0}\right)$. Then $\alpha = -\frac{1}{12}Ax^{-1} \Rightarrow u(\alpha) =$

$$\alpha \sqrt{\left(\frac{u(A)}{A}\right)^2 + \left(-\frac{u(x)}{x}\right)^2}$$

Now $u(A) = \frac{u\left(\frac{N(x)}{N_0}\right)}{\frac{N(x)}{N_0}}$ (Toggerson & Philbin, 2021)

$$\text{But } u\left(\frac{N(x)}{N_0}\right) = u(N(x)N_0^{-1}) = N(x)N_0^{-1} \sqrt{\left(\frac{u(N(x))}{N(x)}\right)^2 + \left(-\frac{u(N_0)}{N_0}\right)^2}$$

Now we get $u(N(x))$ and $u(N_0)$

The uncertainty for a scatter of data is given by: $u(N_{reading}) = \sqrt{\frac{1}{N(N-1)} \sum_{i=1}^N (\bar{N}_{lead} - N_i)^2}$

Using this, we get $u(N_{reading}) = \frac{s(N)}{\sqrt{4}} = 4.38$

Moreover, if we assume the machine is rated at 1%, so $u(N_{rating}) = 0.01N$

$$\text{Now } N_{lead} = \frac{1}{4}N_1 + \frac{1}{4}N_2 + \frac{1}{4}N_3 + \frac{1}{4}N_4$$

$$\text{So } u(N_{rating}) = \sqrt{\left(\frac{1}{4}u(N_{1,rating})\right)^2 + \left(\frac{1}{4}u(N_{2,rating})\right)^2 + \left(\frac{1}{4}u(N_{3,rating})\right)^2 + \left(\frac{1}{4}u(N_{4,rating})\right)^2}$$

$$u(N_{rating}) = \sqrt{\left(\frac{0.86}{4}\right)^2 + \left(\frac{0.67}{4}\right)^2 + \left(\frac{0.84}{4}\right)^2 + \left(\frac{0.75}{4}\right)^2} = 0.392$$

This gives $u(N_{total}) = \sqrt{0.392^2 + 4.38^2} = 4.40$

Table 3: The uncertainty budget for the reading of the lead

An uncertainty budget for the lead block

Uncertainty Component	Standard Uncertainty (Counts)	Type of Evaluation
$u(N_{rating})$	4.38	Type A
$u(N_{reading})$	0.392	Type B
Combined standard uncertainty: $u_c(N) = \sqrt{4.38^2 + 0.392^2} = 4.40$		

Now to get the uncertainty of the counts with no absorber:

The uncertainty for a scatter of data is given by: $u(N_0) = \sqrt{\frac{1}{N(N-1)} \sum_{i=1}^N (\bar{N}_0 - N_i)^2}$

Using this, we get $u(N_0) = 3.75$

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$$\text{So } u\left(\frac{N(x)}{N_0}\right) = \frac{59.2}{197} \sqrt{\left(\frac{4.40}{59.2}\right)^2 + \left(-\frac{3.75}{186}\right)^2} = 0.0231$$

$$\text{Meaning } u(A) = \frac{0.0231}{\frac{59.2}{167}} = 0.0653$$

$$\text{So we can conclude that } u(\alpha) = 0.0864 \sqrt{\left(\frac{0.0653}{\ln\left(\frac{59.2}{167}\right)}\right)^2 + \left(-\frac{0.204}{12}\right)^2} = 0.00563$$

$$\text{So } \alpha = (0.08640 \pm 0.00563) \text{ mm}^{-1}$$

Conclusion and Recommendations

Conclusion

From the Graph analysis, $\alpha = (0.05830 \pm 0.00250) \text{ mm}^{-1} \Rightarrow 0.0558 \leq \alpha \leq 0.0608$

From the Lead block, $\alpha = (0.08640 \pm 0.00563) \text{ mm}^{-1} \Rightarrow 0.0807 \leq \alpha \leq 0.0920$

So the aim has not been met since these intervals do not overlap, they do not agree with each other. It was also obtained that $c = -0.2929 \pm 0.0166$ and by the theoretical equation $-\alpha x = \ln\left(\frac{N(x)}{N_0}\right)$, we need that $c = 0$, so the result does not back the theoretical equation and is thus not reliable.

Discussion and Ways to Improve the Experiment

The Experiment can be improved by:

- Using a more accurate machine
- Providing a means of measuring the thickness in a more accurate way such as thicker slabs

Sources of Uncertainty

- The machine estimating the duration of 60 seconds
- There is a difference in counts for the same distance and same “absorber”. This difference was sometimes very large
- Measuring the thickness of the absorber
- Uncertainty with the rating of the machine

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